OSTBAYERISCHE TECHNISCHE HOCHSCHULE REGENSBURG

ANGEWANDTE NATUR- UND KULTURWISSENSCHAFTEN

THIN GATE OXIDE LAYER ON SEMI- CONDUCTOR

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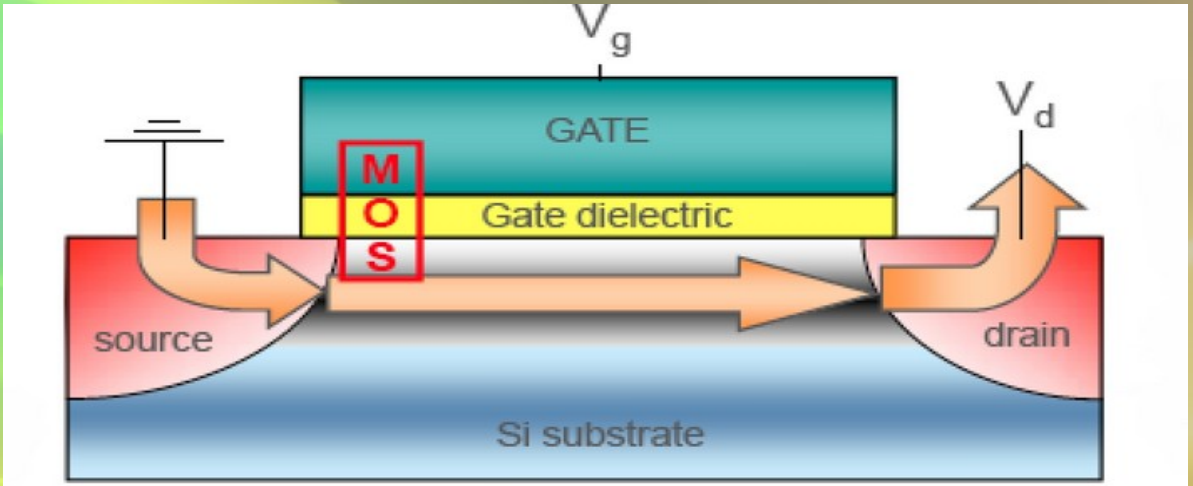
OUTLINE

1. Introduction to Thin Gate Oxide
2. Gate Oxide Importance and Requirements
3. Desired Gate Oxide Characterisations
4. Technology of Thin Oxide growth
5. Nature of Mode-A,B & C of Breakdown failure
6. Oxide strength and charge Breakdown

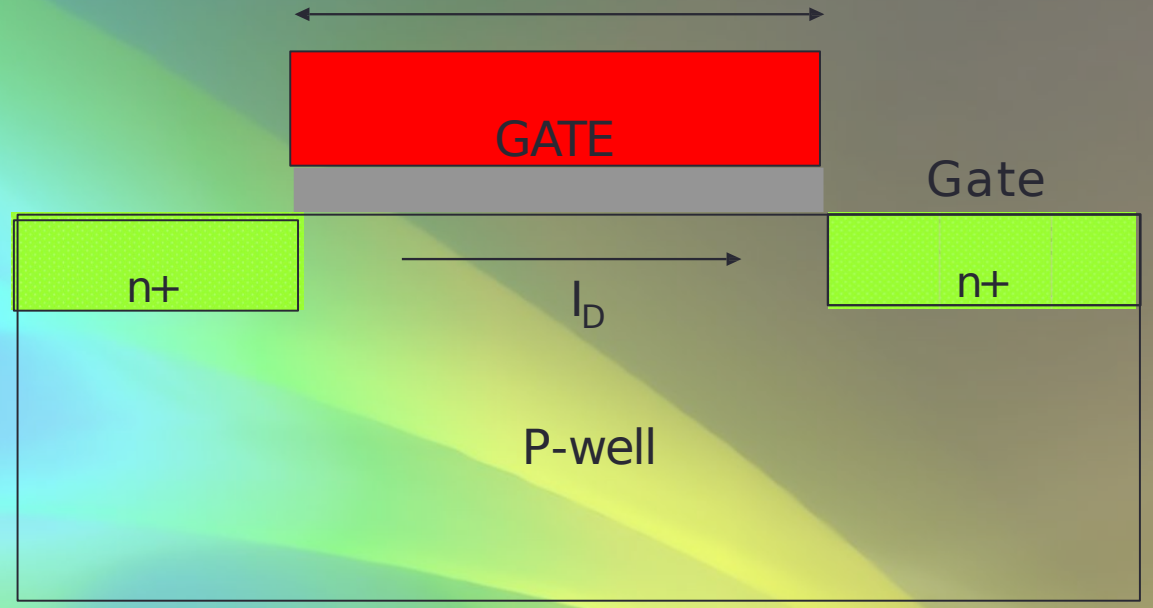
MOSFET

CMOS transistor in digital CMOS circuit is a SWITCH with GAIN.
MOSFET : three terminal device that consists of two pn-junction and one capacitor structure metal/oxide /semiconductor or MOS- Capacitor structure:

- No current flow, only voltage applied source and drain contacts to silicon
- Passing current `under´ the oxide from source to drain (Voltage and current)



MOS : Metal (Poly-Si) – Oxide (SiO_2) – Semiconductor (Si)



$$I_D = \mu_n C_{ox} \frac{W}{L} (V_{gs} - V_{th})^2$$

$$C_{ox} = \frac{\varepsilon A}{t_{ox}}$$

- Gate oxide is composed of Silicon-Dioxide and serves as an insulator between the gate and substrate.
- Gate is main source of Leakage current

GATE OXIDE

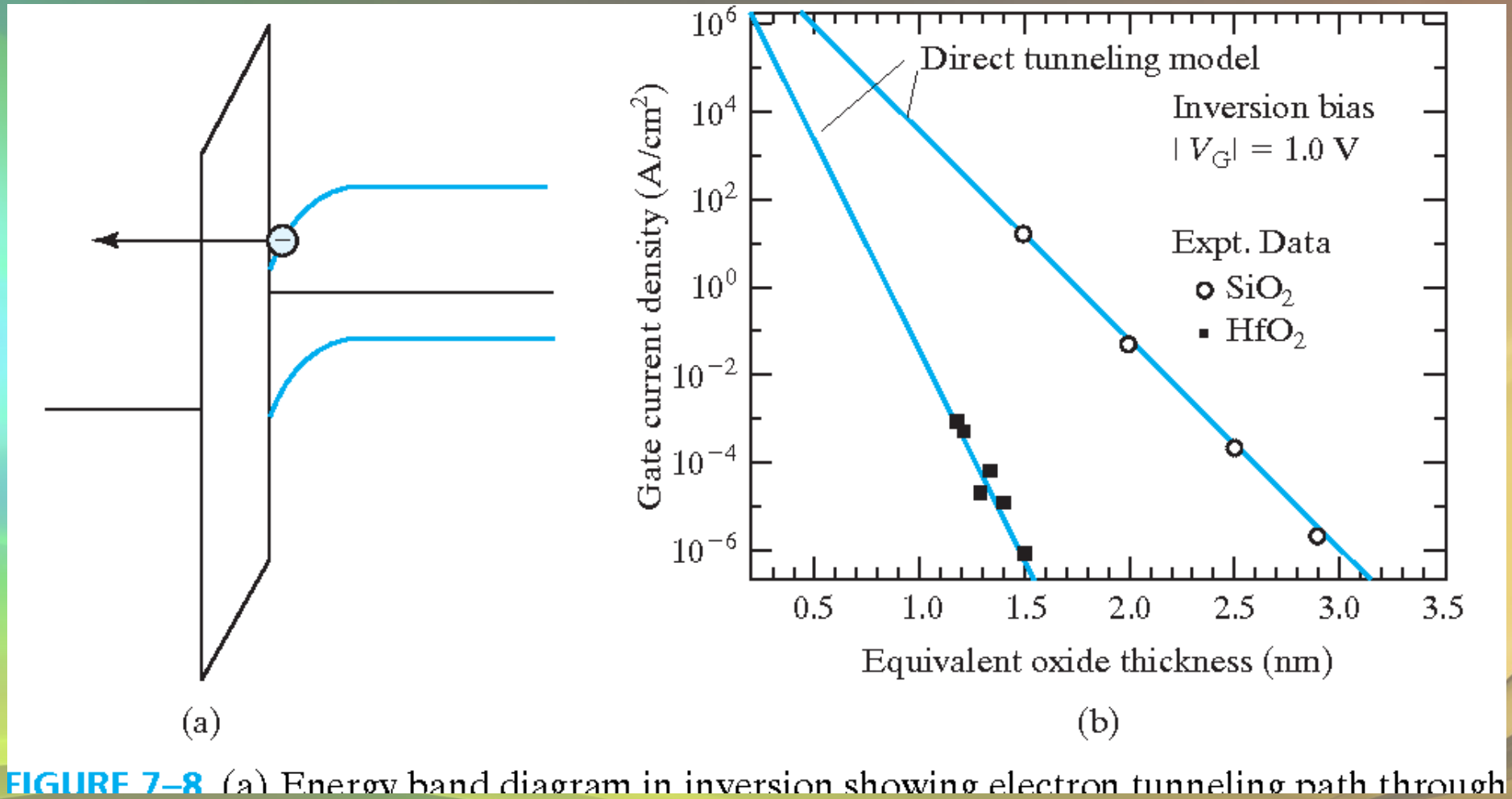


FIGURE 7–8 (a) Energy band diagram in inversion showing electron tunneling path through

Motivation : Why Gate oxide?

Gate Oxide Requirements:

- Possible to grow thin Oxide precisely and uniformly across the wafer
- Adequate reliability characteristics under operating conditions in terms of strength (Breakdown Voltage) , reliability of operation over specified time (τ_{BD}, Q_{BD}) and resistance to hot – carrier degradation.

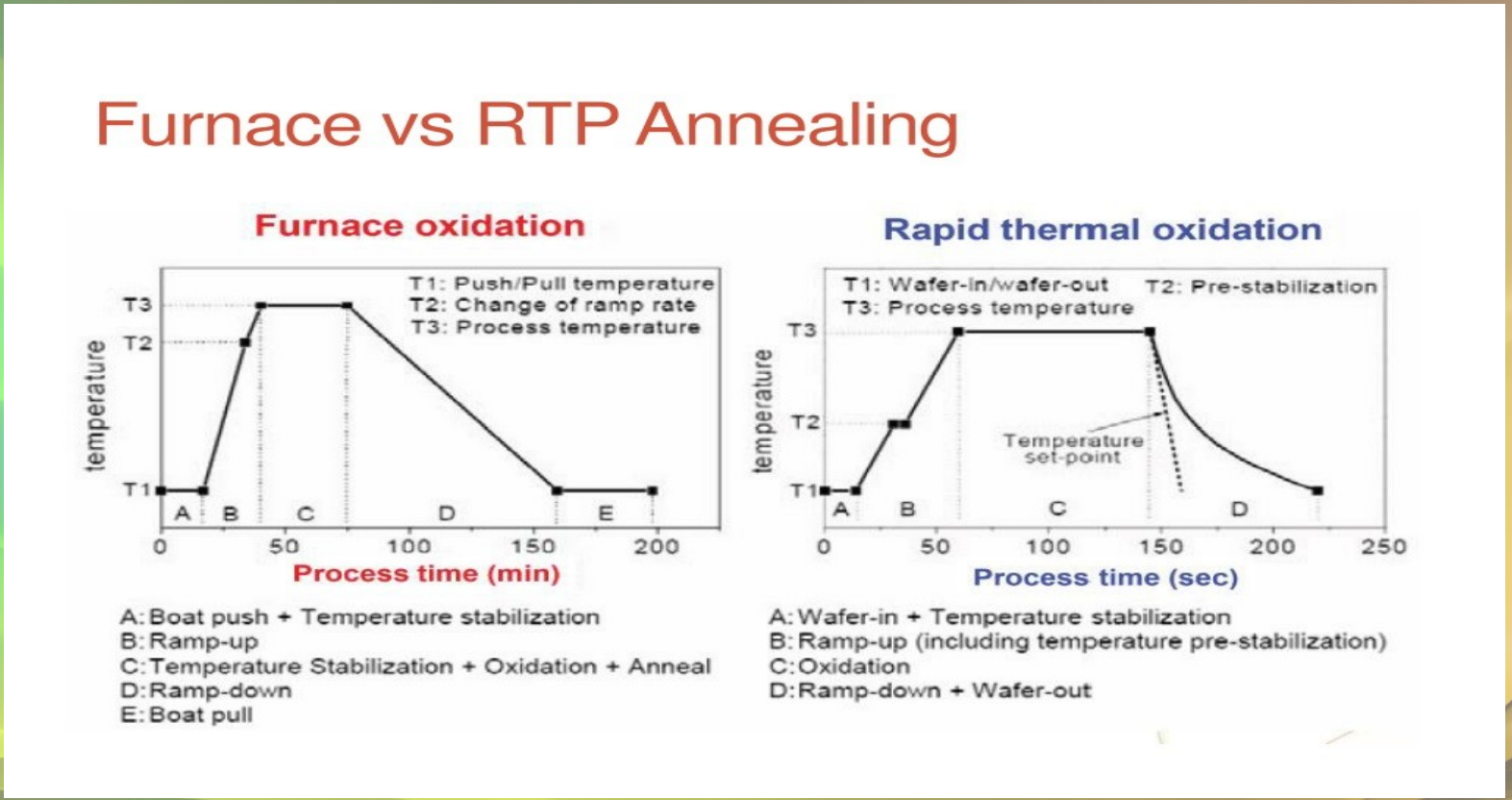
Desired gate Oxide characteristics:

- The thickness closely match the specification in MOSFET design.
- Uniform across the wafer, from wafer to wafer, run to run

- High dielectric strength
- Small interface charge
- Long lifetime under normal operating conditions
- High resistance to hot - carrier damage

Technology of Thin Oxide growth

We have two reliable processes such as Furnance Oxidation and RTP(Rapid thermal process) Annealing.



Technology of Thin Oxide growth

1. Oxidation furnace :

Vertical furnace is more favourable compared to horizontal.
3 reasons:

- Wafer/wafer holders make no contact with oxidation tube during loading, growth and unloading (fewer particles generated).
- Lighter boat material can be used, as a result, better heat and gas distributions (result in better oxide uniformity).
- Precise control of wafer to wafer spacing

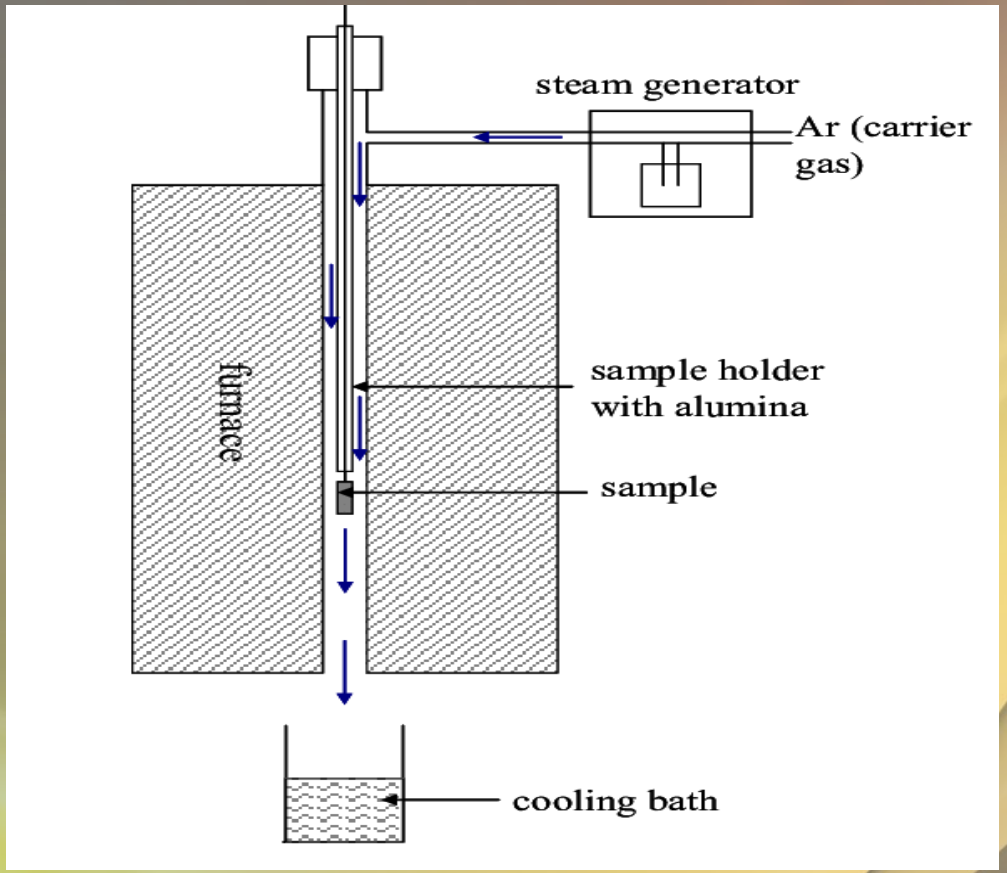
2. Control of Growth Rates:

- Slow growth rates required to reproducibly grow thin oxides with precise thickness.
- Grow in dry O_2 at atmospheric pressure at lower temperatures (800-900C).
- Growth a reduced total pressure, or reduced O_2 partial pressure.

Typical Thin Gate Oxide Process:

1. Carried out in Vertical furnace. Prior to oxide growth.
2. Loaded by robotic wafer handling at specific insertion rates ~ 15 cm per minute for 150mm wafer, 10cm per minute for 200mm wafer. Furnace temperature ~ 650 – 700C.

3. Furnace temperature is ramped up to the growth temperature. During temperature ramp up, N_2 or Ar is purged to prevent unwanted growth.
4. Temperature stabilisation around 5 min before O_2 is released.
5. Furnace temperature is ramped down, and wafer is unloaded at certain rate.

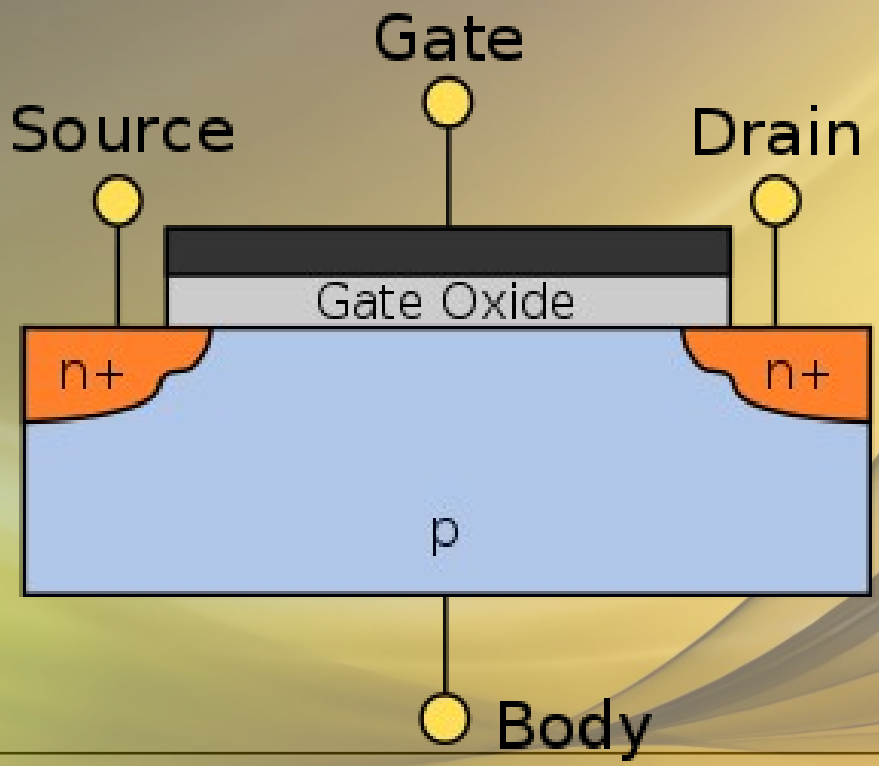


Deposition of Thin Gate Oxide on wafer using Vertical Furnace

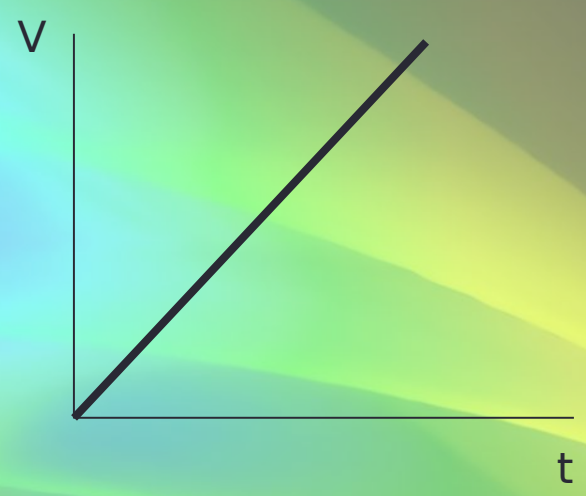
Gate Oxide Characterisations

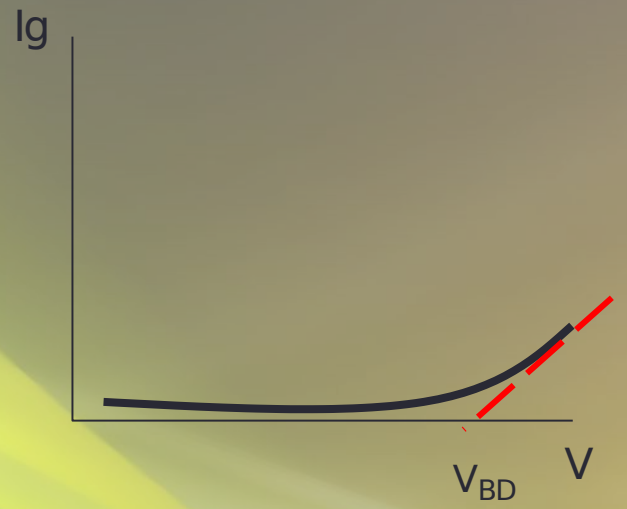
1. Gate oxide Strength
   a) Breakdown voltage, then calculate the breakdown electric field.

2. Gate Oxide Reliability
   a) τ_{BD} – time to breakdown
   b) Q_{BD} – charge to breakdown



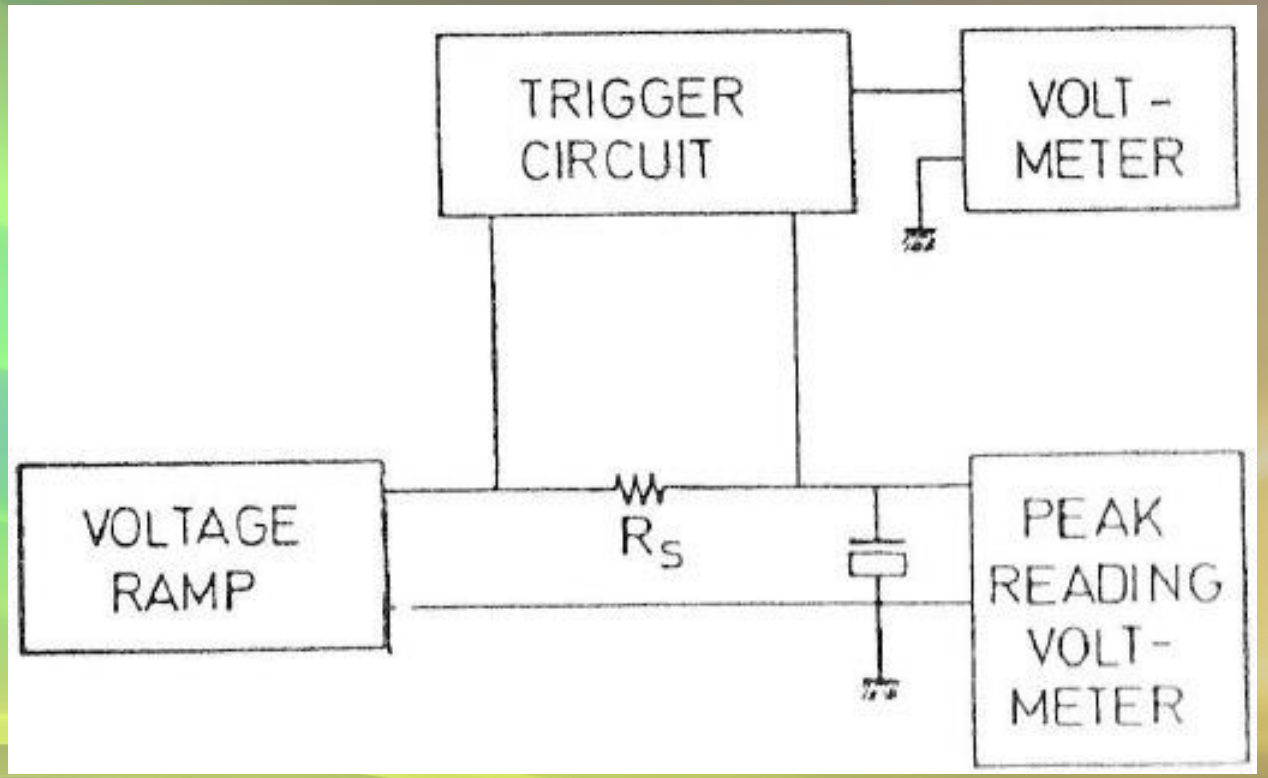
1. Gate Oxide strength:



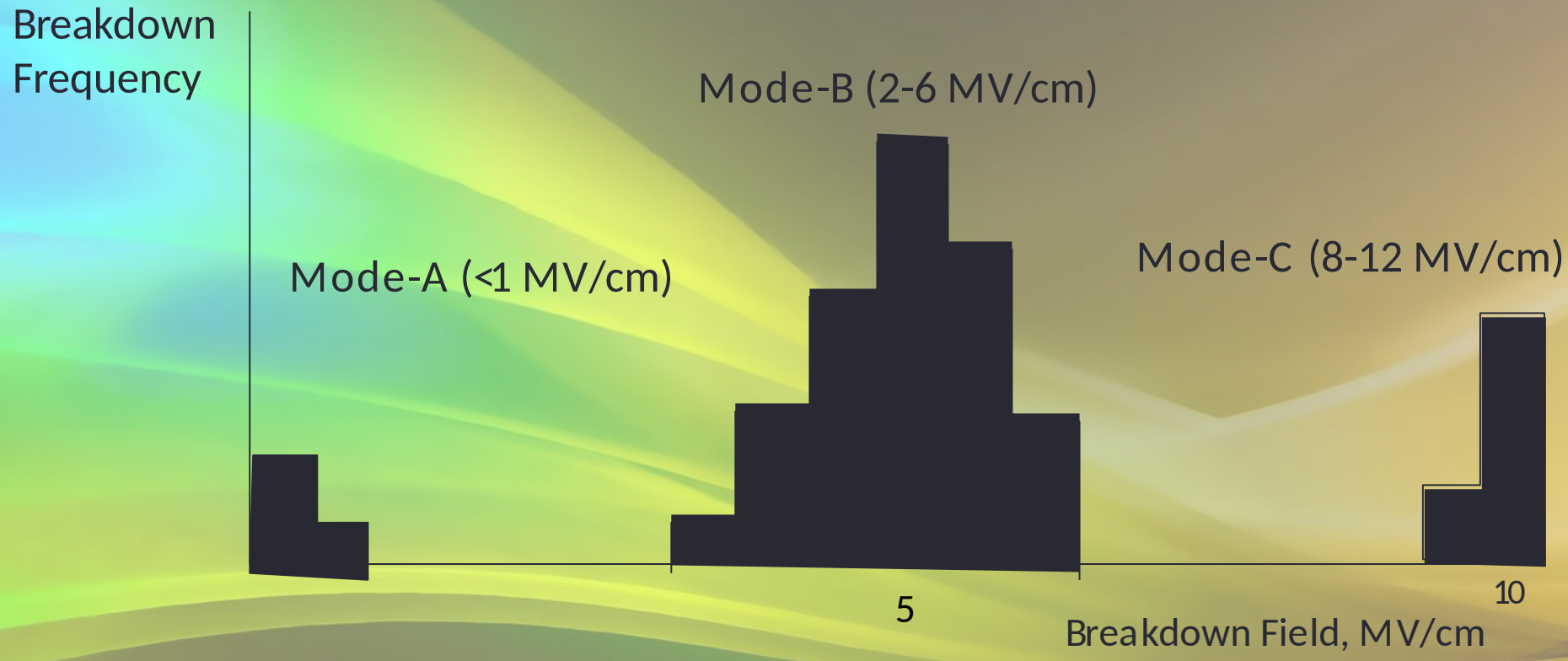


- Take the voltage at the voltage drop as the breakdown voltage (V_{BD})
- Calculate the oxide strength by V_{BD} / oxide thickness

Ramp-Voltage Oxide Breakdown set-up



MODE OF BREAKDOWN FAILURE



2. Gate Oxide Reliability

- Gate oxide strength is not directly relevant to the normal device operation , since what is really needed is how long the thin oxide will survive at lower field strength.
- The measurement of oxide performance at lower electric field called **Time Dependent Dielectric breakdown (TDDB).**

a. Time to breakdown under constant- voltage stressing (τ_{BD})
b. Time to Breakdown under constant- current stressing (τ_{BD})

Charge to Breakdown (Q_{BD})

- The product of current density and t_{BD}

$$Q_{BD} = J_{inj} \cdot \tau_{BD}$$

Emperical Model to EstimateOxide Reliability Applicable to Mode-B and Mode-C Failures

1. Q_{BD} for intrinsic oxide is approximately constant for small FN tunneling current.
2. Q_p (hole charge to breakdown) also constant.

$$Q_p = J_p \alpha \tau_{BD}$$

I. https://www.researchgate.net/figure/Vertical-furnace-used-for-the-oxidation-at-high-temperature-under-steam-flow_fig4_237201468

II. https://en.wikipedia.org/wiki/Gate_oxide#:~:text=The%20gate%20oxide%20is%20the,the%20transistor%20is%20turned%20on

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III. https://en.wikipedia.org/wiki/Field-effect_transistor

IV. https://ieeexplore.ieee.org/document/9955295

Thank You